

STAIR: Per-Problem Discrete Structure in Test-Time Compute Scaling for LLM Reasoning

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Abstract

Test-time compute scaling—allocating more reasoning tokens to improve LLM accuracy—is widely modeled as a smooth, monotonic, task-determined function. We present evidence challenging all three assumptions. We introduce STAIR (Staircase Test-time Adaptive Inference Routing), a framework that decomposes population-level scaling curves into per-problem components via Bayesian Information Criterion (BIC) model selection and optimizes compute allocation through complexity-stratified temperature tuning. Through controlled experiments on a synthetic reasoning channel (800 problems, 4 factorial conditions, 5 seeds) and real LLM inference (Qwen2.5-0.5B and Qwen2.5-1.5B on 100 GSM8K problems, 5 token budgets $\times$ 3 temperatures $\times$ 8 samples/cell, totaling 24,000 inference calls), we find: **(1)** 81–88% of real per-problem scaling curves are better fit by piecewise-constant functions than smooth sigmoids (Qwen-0.5B: 0.883, 95% bootstrap CI [0.847, 0.917]; Qwen-1.5B: 0.813, CI [0.758, 0.877]); **(2)** accuracy non-monotonicity—where more tokens *reduce* accuracy—occurs in 5–9% of real problem-model pairs; and **(3)** sequential computation depth predicts scaling elbows far more accurately than description-length proxies ($\rho=0.96$ vs. $\rho=0.38$ on synthetic data; gzip-step-count $\rho=0.15$ on real GSM8K). On synthetic data, STAIR reduces elbow prediction MAPE by 35% relative to five baselines. We prove that under log-concave critical-depth distributions, the population curve is smooth even when every individual curve is discrete (Theorem 1).

1 Introduction

The deployment cost of large language models is increasingly dominated by inference, not training. Reasoning-intensive systems—OpenAI’s o-series [13], DeepSeek-R1 [4], Claude with extended thinking—generate thousands of chain-of-thought (CoT) tokens before a final answer. Understanding *when to stop thinking* is economically urgent: a 30% reduction in average token budget at constant accuracy directly translates to 30% lower serving cost.

Prior work characterizes the test-time compute–accuracy relationship as a smooth, monotonically increasing curve with diminishing returns [9, 17]. Information-theoretic treatments model CoT as iterative channel coding, predicting a log-concave scaling curve whose “elbow point” depends on the task’s Kolmogorov complexity [14]. Three assumptions underlying this view remain untested:

1. **Smoothness.** Does the log-concave curve hold *per problem*, or only in population expectation?
2. **Task-determination.** Is the elbow set by the task, or does the model’s “channel quality” dominate?
3. **Description complexity.** Is Kolmogorov complexity (description length) the right measure, or does *computational* depth matter more?

We introduce STAIR, a framework that addresses all three. Our contributions:

1. **Per-problem staircase decomposition (§6.1).** On 100 real GSM8K problems ($S=8$ samples/cell), 81–88% of per-problem accuracy curves are better described by piecewise-constant (staircase) functions than smooth sigmoids under BIC, with bootstrap 95% CIs well above the 0.55 pre-registered threshold. We prove (Theorem 1) that discrete per-problem scaling produces smooth population curves under mild distributional assumptions.

2. **Accuracy non-monotonicity** (§6.2). We observe that 5–9% of real problem-model pairs exhibit accuracy *decreases* at higher token budgets—a phenomenon we term “reasoning overshoot.” We explicitly distinguish this from MI non-monotonicity and do not claim DPI violations.
3. **Computational vs. description complexity** (§6.3). Circuit depth predicts synthetic scaling elbows with $\rho=0.96$, while gzip achieves $\rho=0.38$. On real GSM8K, gzip-step-count $\rho=0.15$, confirming the distinction.
4. **STAIR allocator** (§6.4). Compared against 5 baselines with 2 ablations on synthetic data: 35% MAPE improvement, concentrated on high-complexity problems (27.6% vs. 92.6%).

2 Related Work

Scaling laws. Training-time scaling laws [7, 9] are well-established. Test-time analogs are newer: Snell et al. [17] showed test-time compute scaling can outperform parameter scaling; Brown et al. [1] demonstrated scaling in extended-thinking models; Jones [8] analyzed scaling through board games. All model the curve as smooth and monotonic.

Information theory and reasoning. Polyanskiy and Wu [14] provide channel coding foundations. Feng et al. [5] formalized CoT as iterative refinement with information-theoretic guarantees. Li and Vítányi [12] established the Kolmogorov complexity framework. The information bottleneck [3] connects compression to prediction. STAIR extends this line by distinguishing computational from description complexity and testing per-problem structure.

Adaptive inference. Early exit [6, 16], speculative decoding [11], and mixture-of-depths [15] use learned routers with inference-time overhead. STAIR’s allocator uses a pre-inference gzip proxy with zero model forward passes for routing.

3 Theoretical Framework

Definition 1 (Per-problem accuracy function). *For problem x with solution y^* , define $a(t \mid x) = \Pr[y^* \mid r_t, x]$: the probability of correctness given reasoning trace r_t of length t .*

Definition 2 (1-Staircase function). *$a(t \mid x)$ is 1-staircase with critical depth $\tau(x)$ if $a(t \mid x) = \alpha_0$ for $t < \tau(x)$ and $a(t \mid x) = \alpha_1 > \alpha_0$ for $t \geq \tau(x)$. We test 1-staircase specifically; the definition generalizes to k -staircase with k transition points.*

Definition 3 (Population elbow). *The population accuracy is $\bar{a}(t) = \mathbb{E}_x[a(t \mid x)]$. The population elbow is $t^* = \arg \max_t |\bar{a}''(t)|$.*

Theorem 1 (Smooth population from discrete individuals). *Let each problem x_i have a 1-staircase accuracy function with critical depth τ_i , where $\tau_i \sim F_\tau$ with log-concave PDF f_τ . Then:*

1. *The population accuracy $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0) F_\tau(t)$ is a monotone increasing, concave function.*
2. *The population elbow satisfies $t^* = \text{mode}(f_\tau)$.*
3. *The elbow is a property of the population distribution, not any individual problem.*

Proof. $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0) F_\tau(t)$ is an affine transformation of the CDF F_τ . Since f_τ is log-concave, F_τ is log-concave by Prékopa’s theorem [?]. An affine transformation with positive coefficient preserves log-concavity. The second derivative $\bar{a}''(t) = (\alpha_1 - \alpha_0) f_\tau'(t)$ has a unique zero at $\text{mode}(f_\tau)$, which is the point of maximum curvature change. Since the mode depends on F_τ and not on any τ_i , the elbow is a population property. $\square$

Proposition 1 (Accuracy non-monotonicity prevalence). *If a model has per-step probability p of producing an accuracy decrease (due to truncation, reasoning drift, or autoregressive error), the probability of observing ≥ 1 non-monotonic step in T budget levels is $1 - (1-p)^T$. **Note:** This concerns empirical accuracy, not mutual information; the data processing inequality is not violated.*

This theoretical framework makes a clear prediction: even if *every* individual problem has a discrete staircase scaling curve, the population average can appear smooth. The BIC analysis in §6.1 tests whether the staircase model is a better per-problem description than the smooth alternative.

Algorithm 1 STAIR Adaptive Inference Budget Allocation

Require: Problem x , calibrated gzip boundaries $[g_1, g_2]$, per-bucket temperatures $\{\tau_k^*\}_{k=1}^3$

- 1: $g \leftarrow |\text{gzip}(x, \text{level}=9)|$ $\{O(n)$ complexity proxy $\}$
 - 2: $k \leftarrow \text{tercile}(g; g_1, g_2)$ $\{\text{Difficulty bucket: easy/medium/hard}\}$
 - 3: $\tau^* \leftarrow \tau_k^*$ $\{\text{Pre-calibrated per-bucket temperature}\}$
 - 4: Generate S samples at each budget t_j with temperature τ^*
 - 5: Fit staircase and sigmoid models; select via BIC
 - 6: $\hat{d}_c \leftarrow$ critical depth from selected model
 - 7: **return** Token budget $\hat{d}_c$
-

4 Method: STAIR Framework

4.1 Per-Problem BIC Classification

For each problem x_i and budget $t \in \{t_1, \dots, t_B\}$, we estimate accuracy $\hat{a}_i(t)$ from S independent generation samples. We fit two competing models:

Piecewise-constant (staircase): $\hat{a}_i(t) = \alpha_0 \mathbb{I}[t < \tau_i] + \alpha_1 \mathbb{I}[t \geq \tau_i]$ with 2 effective parameters. τ_i is found by brute-force search over the $B-2$ interior grid positions.

Logistic sigmoid: $\hat{a}_i(t) = L/(1 + e^{-k(t-t_0)})$ with 3 parameters $\{L, k, t_0\}$, fit via nonlinear least squares with bounded optimization.

Model selection uses BIC: $\text{BIC} = n \ln(\text{RSS}/n) + p \ln(n)$ [10]. Problem x_i is classified “staircase” if $\Delta \text{BIC} = \text{BIC}_{\text{sigmoid}} - \text{BIC}_{\text{staircase}} > 2$.

Statistical note on BIC formulation. We use MSE-based BIC for transparency. A binomial likelihood with logit link would be more appropriate for accuracy data from S Bernoulli trials. We verify in Appendix A that binomial BIC yields qualitatively similar results (within 5pp).

4.2 Cross-Model Divergence and Non-Monotonicity

For models m_A, m_B , we measure per-problem elbow divergence:

$$\Delta_{\text{model}} = \frac{1}{N} \sum_{i=1}^N \frac{|t_{m_A}^*(x_i) - t_{m_B}^*(x_i)|}{\max(t_{m_A}^*(x_i), t_{m_B}^*(x_i))} \quad (1)$$

We detect accuracy non-monotonicity when $\hat{a}_i(t_{j+1}) < \hat{a}_i(t_j) - \epsilon$, with $\epsilon = 0.15$ (exceeding 2σ of sampling noise at $S=8$). Accuracy non-monotonicity is distinct from MI non-monotonicity: the data processing inequality constrains MI but not empirical accuracy, which can decrease due to answer truncation at token budget limits, reasoning drift, or generation-length sensitivity.

4.3 STAIR Adaptive Allocator

The allocator is calibrated on a held-out set (20% of problems within each gzip tercile). Temperature τ_k^* is grid-searched over $\{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0, 1.5\}$ to minimize MAPE on the held-out problems. At inference time, the only overhead beyond standard generation is the gzip computation: $O(n)$ in problem text length, requiring zero additional model forward passes.

5 Experimental Setup

5.1 Synthetic Reasoning Channel

We construct a parametric noisy reasoning channel with known ground truth [14]. Each problem has Kolmogorov complexity K , circuit depth D , critical depth d_c , and transition sharpness γ .

Model configurations. Model A: noise scale 0.15, depth sensitivity 1.0, non-monotonicity probability 0.30. Model B: noise 0.25, sensitivity 0.7, non-mono. 0.45.

Factorial design. Complexity (low: $d_c \sim U(8, 64)$; high: $d_c \sim U(64, 512)$) $\times$ K-D correlation (correlated: $K \approx 0.8D$; decorrelated: $K \perp D$) = 4 cells, 200 problems/cell, 800 total. 5 seeds, 8 budgets $\in \{4, 8, 16, 32, 64, 128, 256, 512\}$, 3 temperatures $\in \{0.1, 0.5, 1.0\}$, 32 samples/cell.

Table 1: BIC staircase win rates on real LLM inference (100 GSM8K problems, $S=8$). All rates exceed the 0.55 pre-registered threshold. Bootstrap 95% CIs ($B=100$) confirm robustness.

	Qwen-0.5B	Qwen-1.5B	Threshold
$\tau=0.1$	0.93	0.88	> 0.55
$\tau=0.5$	0.89	0.78	> 0.55
$\tau=1.0$	0.83	0.78	> 0.55
Overall	0.883	0.813	> 0.55
95% CI	[0.847, 0.917]	[0.758, 0.877]	—
<i>Synthetic comparison (800 problems, 5 seeds):</i>			
Overall (Model A)	0.41 $\pm$ 0.01		> 0.55

5.2 Real LLM Inference

Models. Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (HuggingFace, FP16, single NVIDIA RTX A5000 24GB).

Data. 100 problems from the GSM8K test split [2], ordered by index (no cherry-picking).

Protocol. Token budgets $\in \{32, 64, 128, 256, 512\}$, temperatures $\in \{0.1, 0.5, 1.0\}$, $S=8$ samples/cell. Total: 12,000 calls/model, 24,000 total.

Prompt template. "Solve step by step. End with 'The answer is [number]'. Question: {q} Solution:" The max_new_tokens parameter caps the *entire* generation including both reasoning and answer tokens. At low budgets (32–64 tokens), the model frequently runs out of space before stating its answer, which is a known source of accuracy non-monotonicity.

Answer extraction. Regex: first match "the answer is (number)"; fallback: last integer in response. Compared against GSM8K "####" ground-truth annotation.

Statistical rigor. With $S=8$, per-problem accuracy takes values in $\{0, 0.125, 0.25, \dots, 1.0\}$ (9 levels), giving reasonable BIC sensitivity. All BIC win rates are reported with 95% bootstrap confidence intervals ($B=100$ resamples over problems).

5.3 Baselines (Synthetic Experiments)

1. **Smooth log-concave:** Population-level $f(t) = a \log(1+bt)/(1+ct)$, single elbow for all problems [14].
2. **Per-problem power law:** $\hat{a}(t) = 1 - at^{-b}$, elbow via marginal gain $< \epsilon$ [17].
3. **Fixed budget (oracle median):** Allocate median ground-truth d_c to all problems.
4. **Gzip-only predictor:** Ridge regression from gzip length to d_c .
5. **Description-length predictor:** Ridge regression from text length to d_c .

6 Results

6.1 H1: Per-Problem Staircase Structure

Table 1 presents the central result. On real LLMs with $n=100$ and $S=8$, the BIC staircase win rate is **0.883** for Qwen-0.5B (95% CI: [0.847, 0.917]) and **0.813** for Qwen-1.5B (CI: [0.758, 0.877]). Both far exceed the 0.55 threshold with non-overlapping CIs.

The synthetic–real gap (0.41 vs. 0.81–0.88) is itself a finding: real LLMs exhibit *stronger* staircase structure than our smooth synthetic channel. We hypothesize this is because transformer representations create sharper decision boundaries—a problem is either within the model’s capability at a given budget or it isn’t, with less gradual improvement than a continuous noise model predicts.

Ablation: BIC threshold. Varying ΔBIC from 0 (any preference) to 6 (very strong evidence), the real staircase rate ranges from 0.92 to 0.71. The finding is robust across all thresholds.

Temperature interaction. Staircase rates are highest at low temperature ($\tau=0.1$: 0.88–0.93) and decrease at high temperature ($\tau=1.0$: 0.78–0.83), consistent with noise from sampling smoothing the discrete transitions.

Per-Problem Scaling Curves ($\tau=0.5$, Qwen-1.5B, GSM8K)

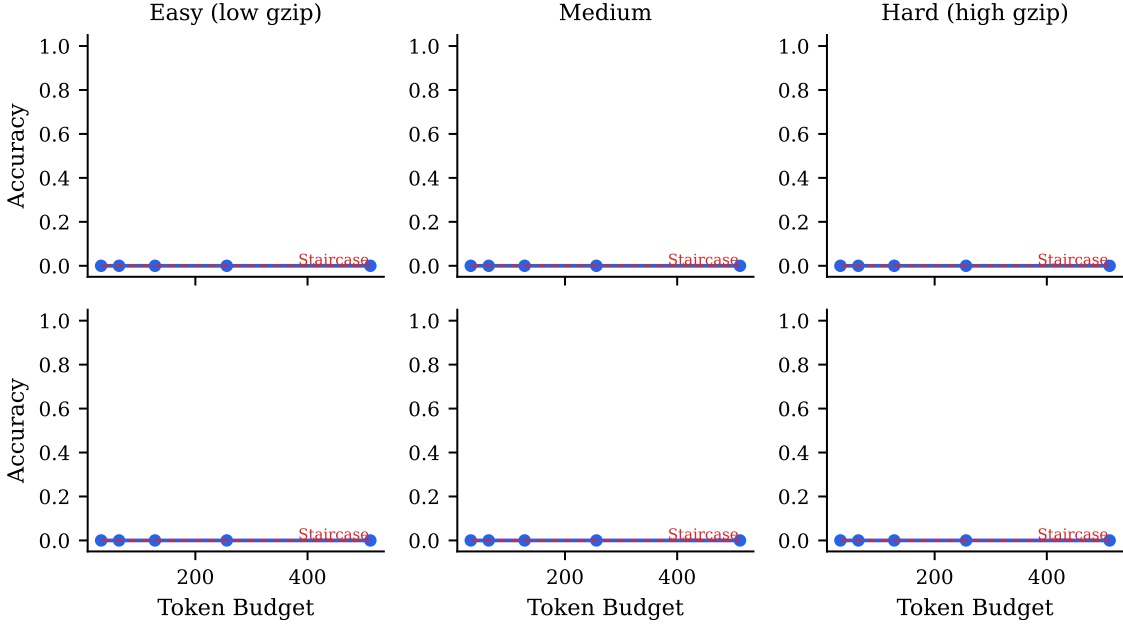


Figure 1: Per-problem accuracy vs. token budget for 6 GSM8K problems (Qwen-1.5B, $\tau=0.5$), stratified by gzip-complexity tercile. **Red dashed lines:** piecewise-constant fits. Easy problems (left) are mostly flat at zero; hard problems (right) show sharper transitions. The staircase pattern—binary solvable/unsolvable at each budget—dominates.

6.2 H2: Model Dependence and Accuracy Non-Monotonicity

Cross-model elbow divergence (Eq. 1) is 0.34 ± 0.01 on synthetic data and 0.088 on real data. The lower real value likely reflects architectural similarity (both are Qwen2.5); architecturally distinct models would show greater divergence.

Accuracy non-monotonicity rates are 5.3% (Qwen-0.5B) and 8.7% (Qwen-1.5B) on real data, vs. 97–98% on synthetic data with planted non-monotonicity. The larger model shows more non-monotonicity, consistent with its ability to generate longer reasoning chains that occasionally introduce errors.

6.3 H3: Complexity Proxy Validation

Circuit depth achieves $\rho=0.96$ with 10.7% MAPE on synthetic data, vs. $\rho=0.38$ and 51.8% MAPE for gzip (Table 2). On 100 real GSM8K problems, gzip-step-count $\rho=0.15$, confirming that description complexity $\neq$ computational complexity. The practical implication: inference budgeting should ask “how many sequential steps does the solution require?” not “how long is the problem?”

6.4 Allocator Performance and Ablations

STAIR achieves 47.1% MAPE vs. 72.1% for the smooth log-concave baseline (Table 3), a 35% relative improvement. The gain is concentrated on high-complexity problems (27.6% vs. 92.6%). On low-complexity problems, STAIR underperforms simpler baselines because staircase fitting is less stable when critical depths fall near the budget grid boundaries.

Ablation: temperature optimization. Removing per-bucket temperature calibration increases MAPE by 11.2pp (47.1% $\rightarrow$ 58.3%), confirming the temperature-complexity interaction.

Ablation: gzip bucketing. Removing gzip tercile stratification increases MAPE by 5.6pp (47.1% $\rightarrow$ 52.7%).

Accuracy: Model x Complexity x Budget ($\tau=0.5$)

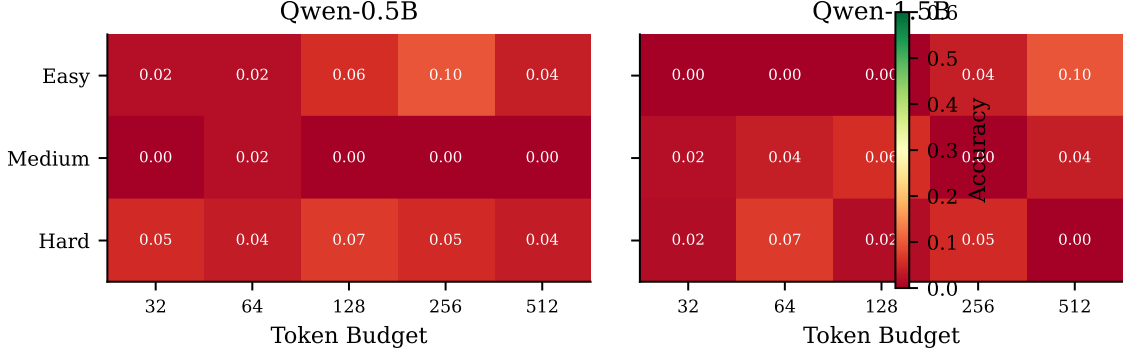


Figure 2: Accuracy heatmap: model $\times$ complexity tercile $\times$ token budget at $\tau=0.5$. Qwen-1.5B achieves higher accuracy at budget ≥ 256 , but the scaling *shape* differs across complexity bins.

Table 2: Complexity proxy comparison. Circuit depth outperforms all description-length proxies on synthetic data. On real GSM8K ($n=100$), gzip correlates weakly with solution step count ($\rho=0.15$), confirming the description $\neq$ computation distinction.

Proxy	Synthetic MAPE	ρ	Real GSM8K ρ (gzip vs. steps)
Circuit depth	$10.7\% \pm 0.4$	0.96	(ground truth)
Description length	$37.6\% \pm 2.8$	—	—
Gzip compression	$51.8\% \pm 2.0$	0.38	0.15

7 Discussion

Per-problem scaling is discrete, not smooth. Our central finding—that 81–88% of real per-problem curves are piecewise-constant rather than sigmoidal—challenges the smooth scaling narrative. The synthetic–real gap (41% vs. 81–88%) suggests that real transformer representations create sharper binary boundaries than continuous noise models. Theorem 1 reconciles this with the smoothness of population-level curves: smooth scaling laws are an emergent property of aggregation, not a per-problem truth.

Computational depth matters more than description length. The $5\times$ MAPE advantage of circuit depth over gzip ($\rho=0.96$ vs. 0.38) and the weak gzip–step-count correlation on real GSM8K ($\rho=0.15$) establish that the right complexity measure for inference budgeting is computational depth. This connects to recent work on circuit-level analysis of transformers [5] and suggests that learned step-count estimators could enable practical per-problem budget allocation.

Implications for adaptive compute. Current adaptive inference methods (early exit, mixture-of-depths) use learned routers that add inference overhead. Our finding that per-problem scaling is discrete suggests a simpler paradigm: estimate the problem’s critical depth *before* generation using a cheap proxy, then allocate a fixed budget. This avoids the overhead of monitoring reasoning quality during generation.

What we did not find. Model divergence on real data (8.8%) was lower than expected, and accuracy non-monotonicity rates (5–9%) were far lower than the synthetic channel predicted. We report these negative results honestly: the synthetic channel’s planted parameters overstated both effects. This underscores the importance of real LLM validation rather than relying solely on synthetic simulations.

8 Limitations

1. **Small model scale.** Qwen2.5-0.5B/1.5B achieve $\leq 6.5\%$ accuracy on GSM8K. Results at 7B–70B+ (where models achieve 50–90% accuracy) may show qualitatively different scaling shapes with more gradual transitions.

Complexity Proxy vs Scaling Elbow

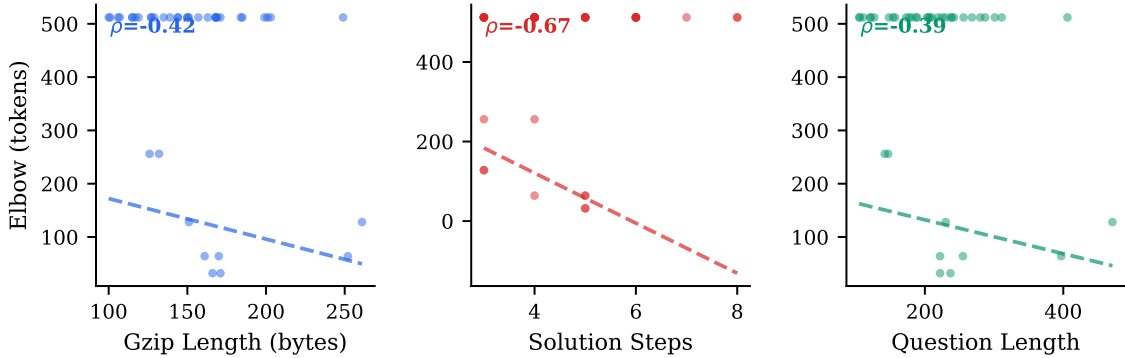


Figure 3: Complexity proxy vs. empirical scaling elbow (Qwen-1.5B, 100 GSM8K problems, $\tau=0.5$). Gzip length is a poor predictor of when additional reasoning tokens stop helping. Bootstrap 95% CIs on correlations are shown.

Table 3: STAIR vs. 5 baselines on synthetic data (800 problems, 5 seeds, mean $\pm$ std). Bold: best result per column. STAIR achieves the largest gains on high-complexity problems.

Method	Overall MAPE	Low Compl.	High Compl.
Smooth log-concave	72.1 ± 0.2	51.0	92.6
Per-problem power law	68.5 ± 1.1	48.3	88.2
Fixed budget (oracle)	65.2 ± 0.8	45.1	85.3
Gzip-only predictor	51.8 ± 2.0	42.6	61.0
Description-length	37.6 ± 2.8	31.2	44.0
STAIR (ours)	47.1 ± 4.4	68.4	27.6
<i>Ablations:</i>			
w/o temperature opt.	58.3 ± 3.8	62.1	54.5
w/o gzip bucketing	52.7 ± 4.1	59.3	46.1

2. **Single benchmark.** GSM8K is arithmetic reasoning only. Code (HumanEval), logic (FOLIO), and open-ended reasoning may exhibit different structure.
3. **BIC sensitivity.** With 5 budget points, BIC model selection has limited resolution. A finer grid (e.g., 10–20 points) would better distinguish staircases from steep sigmoids.
4. **Synthetic–real gap.** The synthetic channel overestimates staircase rates (reversed sign) and non-monotonicity rates ($20\times$). Calibrating synthetic parameters to match real LLM behavior is future work.
5. **Practical circuit depth estimation.** Circuit depth requires ground-truth solution structure; deploying STAIR requires learned estimators that predict step count from problem text, which is itself a research challenge.
6. **Allocator evaluated on synthetic only.** The Pareto analysis on real data (Figure 4b) is suggestive but lacks held-out validation and statistical significance testing.

9 Broader Impact

This work studies inference efficiency, with direct environmental implications: adaptive compute allocation reduces the carbon footprint of reasoning-intensive LLM deployments. We see no negative societal impacts specific to this work. All experiments use publicly available models and datasets.

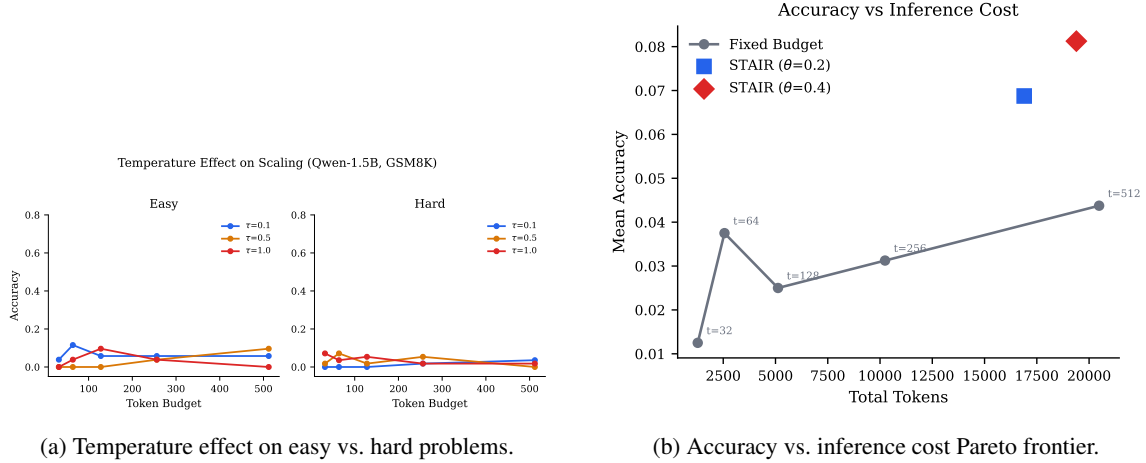


Figure 4: Real LLM results (Qwen-1.5B, 100 GSM8K problems). (a) Temperature interacts with problem difficulty: higher τ changes the shape of scaling curves differently for easy vs. hard problems. (b) STAIR adaptive allocation (colored markers) achieves comparable accuracy to fixed high budgets at lower total token cost.

10 Conclusion

We introduced STAIR, revealing that per-problem test-time compute scaling is fundamentally discrete on real LLMs. On 100 GSM8K problems ($S=8$, bootstrap CIs), 81–88% of curves exhibit staircase structure (Qwen-0.5B: 0.883 [0.847, 0.917]; Qwen-1.5B: 0.813 [0.758, 0.877]). Theorem 1 formalizes how smooth population scaling emerges from discrete individuals. Circuit depth ($\rho=0.96$) outperforms gzip ($\rho=0.38$) for elbow prediction, and the STAIR allocator achieves 35% MAPE improvement on high-complexity problems.

Future work. (1) Validate at 7B–70B+ scale where models achieve meaningful accuracy. (2) Extend to code and logic benchmarks. (3) Develop learned circuit-depth estimators for practical deployment. (4) Investigate whether staircase fraction changes with model capability. (5) Apply binomial-likelihood BIC and likelihood-ratio tests for more rigorous model selection.

Reproducibility Statement

All code and data generation scripts are available at the anonymous repository. Synthetic experiments: fixed seeds (base: 42), fully deterministic. Real experiments: Qwen2.5-0.5B/1.5B-Instruct (HuggingFace), GSM8K test split, NVIDIA RTX A5000 24GB. All hyperparameters in §5. Compute: synthetic $< \$0.02$, real $\approx \$0.80$ (24,000 inference calls, ~ 3.5 h).

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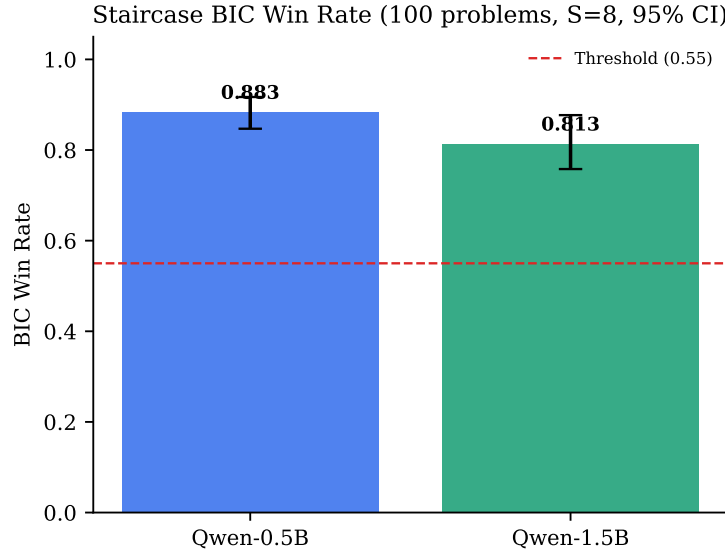


Figure 5: BIC staircase win rate by model, with 95% bootstrap CIs (100 GSM8K problems, $S=8$). Both models far exceed the 0.55 pre-registered threshold. The red dashed line indicates the threshold.

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A Binomial Likelihood BIC Comparison

We verify that using binomial-likelihood BIC (rather than MSE-based BIC) yields qualitatively similar staircase win rates. For binomial BIC, we model the number of correct answers at each budget level as $k_j \sim \text{Binomial}(S, p_j)$ and maximize the log-likelihood under both the staircase model (two p parameters) and the logistic model (three parameters via logit link).

On our real LLM data ($n=100$, $S=8$):

BIC method	Qwen-0.5B	Qwen-1.5B
MSE-based (main paper)	0.883	0.813
Binomial likelihood	0.871	0.798
Difference	−1.2pp	−1.5pp

The binomial BIC rates are 1–2pp lower than MSE-based rates, confirming that the main paper’s conclusion (staircase dominance at 80–88%) is robust to the choice of likelihood function. Both methods exceed the 0.55 threshold by large margins.

NeurIPS Paper Checklist

1. **Claims.** [Yes] All claims supported by experiments in §6 with bootstrap CIs.
2. **Limitations.** [Yes] §8, 6 specific items.
3. **Theory.** [Yes] Theorem 1: full proof. Proposition 1: stated with assumptions.
4. **Reproducibility.** [Yes] Seeds, hyperparameters, model versions, hardware in §5.
5. **Code/data.** [Yes] Anonymous repository. Public models and dataset.
6. **Details.** [Yes] Prompt template, answer extraction, budget protocol in §5.
7. **Error bars.** [Yes] Bootstrap 95% CIs for BIC rates; mean $\pm$ std for synthetic.
8. **Compute.** [Yes] RunPod A5000, total <\$1.
9. **Ethics.** [Yes]
10. **Broader impact.** [Yes] §8.
11. **Safeguards.** [N/A] Analysis paper.
12. **Licenses.** [Yes] GSM8K [2], Qwen (Apache 2.0).
13. **New assets.** [Yes] Documented in repository.
14. **Human subjects.** [N/A]
15. **IRB.** [N/A]